

26.02.10

Value Iteration: $V_{k+1}(s) = \max_a \{ R_s^a + \gamma \sum_{s'} P_{ss'}^a V_k(s') \}$

Policy Iteration: (i) Evaluate π_k , $BG \Rightarrow q_{\pi_k}(s, a)$
(ii) Optimize greedily $\pi_{k+1} = \arg \max_a q_{\pi_k}(s, a)$

They converge to the optimal π^* exponentially fast if $\gamma < 1$.

(3) Monte Carlo $\begin{cases} \text{First Visit} \\ \text{Every Visit} \end{cases}$

$\pi: \{ (s_1, R_1), (s_2, R_2), \dots, (s_T, R_T) \}$

$$G_t = R_t + \gamma R_{t+1} + \dots + \gamma^{T-t} R_T$$

$V(s) = \frac{1}{N(s)} \sum_{S \sim s} G_t = \text{Average return across all episodes}$

$$N(s) \leftarrow N(s) + 1$$

$$V(s) \leftarrow V(s) + \frac{1}{N(s)} (G_t - V(s))$$